Western coast offshore oil fields	Eastern coast offshore Fields
1. Mumbai high oil fields (largest 65%)	Bharmaputra valley (Dibrugarh and Sibsagar districts of upper Assam.)
2. Gujarat coast (2nd largest)	Digboi oil feilds (oldest fields in country)
3. Basseim oil feild, south of Mumbai high	Nahoratiya oil fields (south west of digboi)
4. Aliabet oil feild, south of Bhavanagar	Moran-Hugrijan oil field (Southwest of Nahoratiya)
5. Ankleshwar	Rudrasagar-Lawa oil feilds (sibsagar districts of assam)
6. Cambay-Luni Region	Surma valley (Badarpur, Masimpur, Patharia)
7. Ahemedabad-Kalol Region	Offshore of Andaman and Nicobar, Gulf of mannar, Baleshwar coast, Punjab, Haryana and Uttar Pradesh.

Petroleum is used as a source of power and fuel for automobiles, aeroplanes, ships and locomotives. Lubricants, kerosene, vaseline, tar, soap, terylene and wax are its by products. Oil in India is obtained from both from on-shore and off-shore areas.

c) Natural Gas

Natural gas usually accompanies the petroleum accumulations. It is naturally occurring hydro carbon gas mixture consisting primarily of methane, but commonly includes varying amounts of other higher alkanes and sometimes a small percentage of carbon dioxide, nitrogen and hydrogen sulphides. It is formed when layers of decomposed plants and animals are exposed to intense heat and pressure over thousands of years. It is used as a source of energy for heating, cooking and electricity generation. It is also used as fuel for vehicles and as a chemical feedstock in the manufacture of plastics and other commercially important organic chemicals.

GAIL (formerly known as Gas Authority of India Limited) is the largest state-owned natural gas processing and distribution company in India. It is headquartered in New Delhi.

India has a very large proportion of tertiary rock and alluvial deposits particularly in the extra peninsular India. These sedimentary rocks, which were once under the shallow seas, hold the possibility of harbouring oil and gas deposits. The highest concentration of natural gas is found in the Mumbai high and basseim oil fields. Gujarat, Assam, Neypaltur, Mangmadam in Thanjavur district in Tamil nadu, Tripura, Rajasthan, Arunachal Pradesh, Punjab, West Bengal are the other areas where natural gas reserves have been discovered.

Compressed natural gas (CNG) (methane stored at high pressure) is a fuel which can be used in place of gasoline, diesel fuel and propane/LPG. In comparison to other fuels, natural gas poses less of a threat in the event of a spill, because it is lighter than air and disperses quickly when released. Biomethane – cleaned-up biogas from anaerobic digestion or landfills – can be used. Natural gas run vehicles are increasingly used in Delhi, Ahmedabad, Mumbai, Pune, Kolkata Lucknow, Kanpur, Varanasi, etc.

Conventional Energy Sources

a) Thermal power

Thermal power is generated using fossil fuels like coal, diesel, petroleum and Natural

14_Geography_Unit_4_EM.indd 175 23-03-2020 17:46:57